JULES BAJI

R&D ENGINEER · GRAPHICS, GEOMETRY & 3D ML

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Europe



MSc in Computer Science specialising in rendering, geometry and performance for 3D. Machine learning in PyTorch and in native C++/CUDA. Recent experience at CEA Saclay on Gaussian Splatting for VR. Open to international relocation.

SCIENTIFIC & TECHNICAL EXPERIENCE

R&D Internship : Gaussian Splatting

01/2025 – 07/2025 — CEA Saclay

C++ | CUDA | OpenMP | PyTorch | Gaussian Splatting | Ray Tracing | VR

Improved photorealistic Gaussian Splatting rendering for VR, focusing on complex reflections and real-time performance.

- State-of-the-art review and implementation of a ray-tracing prototype.
- Pivoted to a rasterization pipeline integrating Spherical Gaussians to model complex view-dependent reflections, reaching near-ground-truth rendering while preserving real-time performance in VR (+50% storage, +15% training).

Internship : CAD Geometry

04/2023 – 05/2023 — Kreon Technologies

C++ | OpenMP Geometric primitive detection (tori, B-splines) and tessellation algorithm fixes (topological continuity, normal consistency) in production CAD software.

Physically Based Renderer

2025 — University of Limoges

C++ | OpenMP Implemented a path tracer with microfacet BRDFs, SDF and CSG geometry, accelerated by a BVH.

Real-Time 3D Engine

2025 — University of Limoges

C++ | OpenMP | OpenGL | GLSL Built a 3D engine handling meshes, textures, cameras, lights and shaders.

3D Collision Detection

2025 — University of Limoges

WebGL | JavaScript Built a collision system combining Sweep and Prune, BVH and Möller's triangle-triangle test.

Parallel Computing

2025 — University of Limoges

CUDA Implemented and optimised kernels for visibility computation on a height map.

MPI | OpenMP Hybrid implementation of a sub-image search.

Computer Vision

2025 — University of Limoges

Python | Keras Face generation, image super-resolution, and recaptured-photo detection based on a research paper.

Fluid Simulation

2025 — University of Limoges

Java Real-time SPH fluid simulation with grid-based neighbour search, from a research paper.

EDUCATION

M.Sc. Computer Science, Computer Graphics

09/2023 – 09/2025
University of Limoges

B.Sc. Computer Science

09/2020 – 06/2023
University of Limoges

TECHNICAL SKILLS

C++ | CUDA | Python | PyTorch | NumPy | OpenGL | WebGL | GLSL | MPI | OpenMP | Git | JavaScript | Linux | 3D Gaussian Splatting

LANGUAGES

French : Native
English : C1

REFERENCES

Gilles Rougeron

Research engineer, CEA
Contact on request

INTERESTS

- Trumpet and music theory (conservatory, 2011–2019)
- Treasurer of the master's student association
- Climbing